



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal to SREE PALANIYAPPAA
AGENCYIOCDEALERS**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

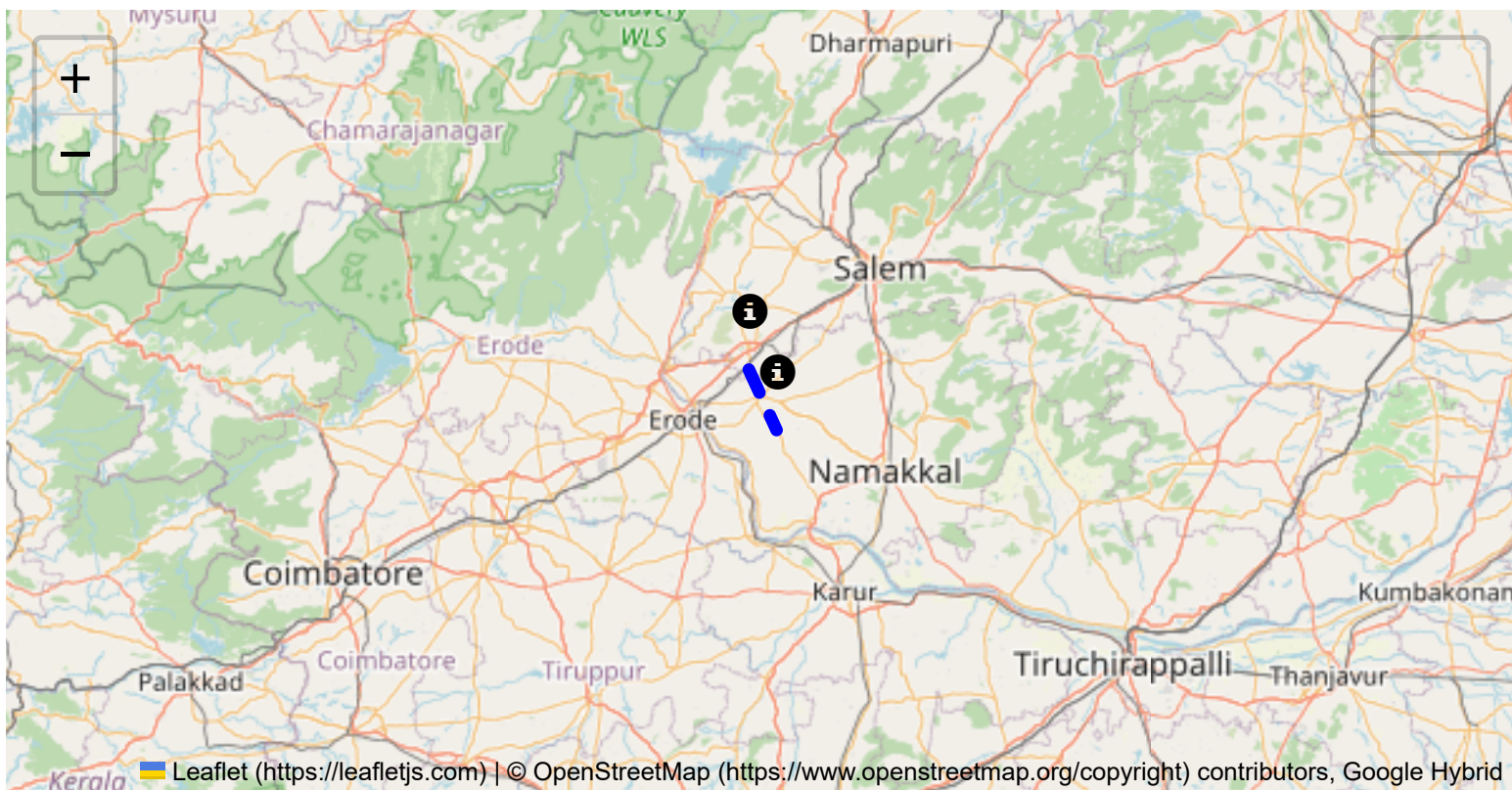
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

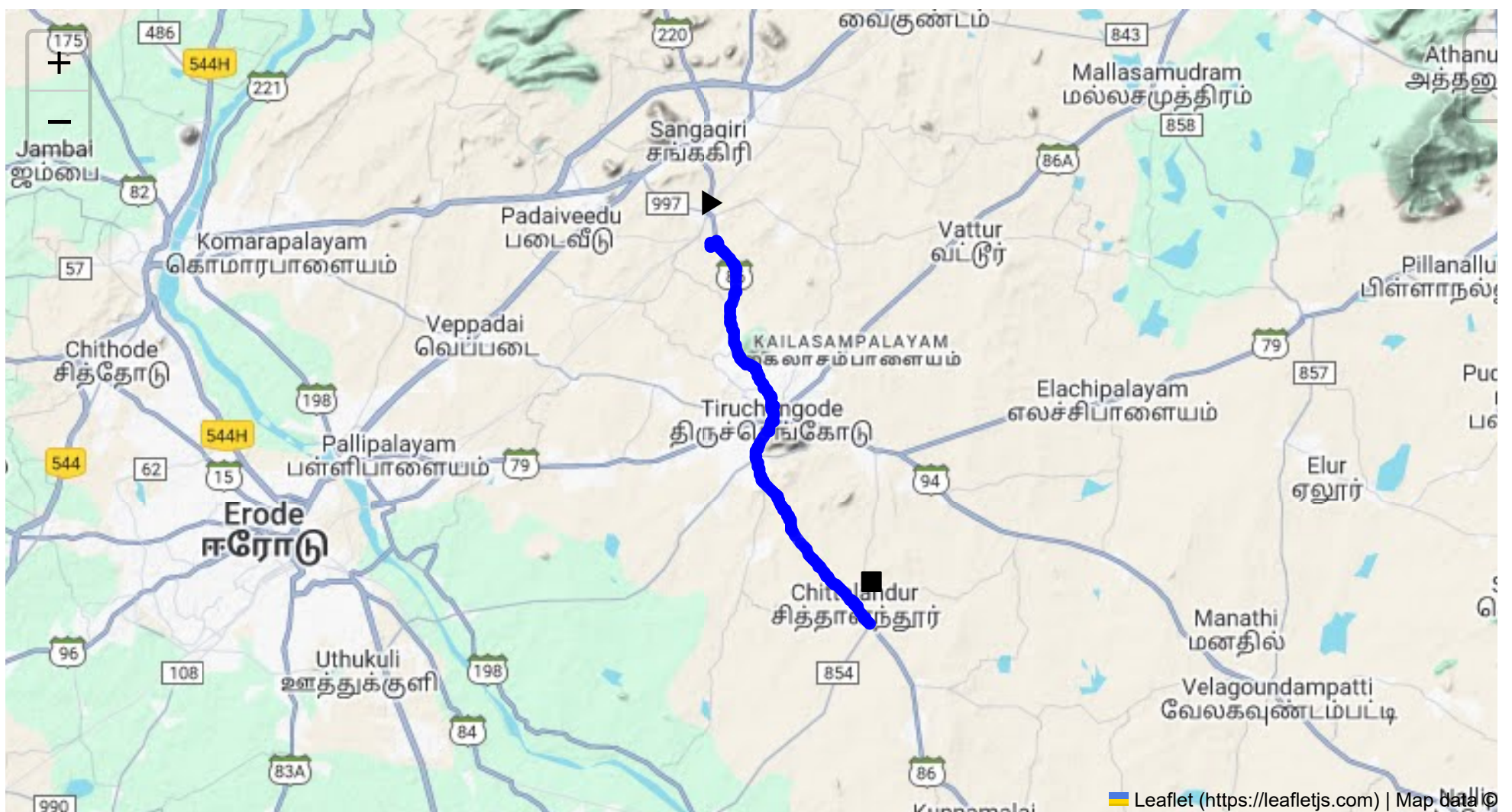
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 17.56 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.308703, 77.928925)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 8W5H+8FW, Sullipalayam spans approximately 17.56 kilometers. This route primarily consists of National Highway segments and rural roads. It typically takes around 0.47 hours for heavy vehicles carrying hazardous material. The area is predominantly rural, with segments passing through small towns and agricultural zones.

Weather Conditions

The region around Tamil Nadu typically experiences a tropical climate. Throughout the year, the area can have substantial rainfall, particularly between October and December due to the northeast monsoon. During heavy rains, the roads can become slippery, increasing the risk of skidding and accidents for heavy vehicles. Heat during the summer may also impact vehicle operations, especially during peak daylight hours.

Traffic Patterns

1. **Peak Hours:** Morning (7:00 AM - 9:00 AM) and Evening (5:00 PM - 7:00 PM) see higher traffic due to local commuters.
2. **Congestion-Prone Areas:** Congestion is likely near towns and marketplaces en route. This can delay travel and increase the risk of accidents during rush hours.

Road Quality and Infrastructure

The road quality on the National Highways section is generally good, but rural and local roads may have patches, potholes, and narrower lanes. Limited street lighting is a concern for night travel. Signage may not always be clear, especially on rural junctions.

Alternative Routes

In case of emergencies, it is advisable to switch to alternate state highways or designated rural roads. Pre-planned emergency detour routes should be identified in advance using GPS navigation tools.

Local Regulations for Hazardous Material Transport

Tamil Nadu requires heavy vehicles carrying hazardous materials to comply with strict regulations, including appropriate marking and permits. Night operations may be restricted, and specific routes may be designated for such transport.

Historical Incidents

There are no major publicized historical incidents involving heavy vehicles or hazardous materials along this specific route. However, typical risks include rollover incidents due to sharp turns on rural roads, especially post-rainfall conditions.

Environmental Considerations

The route passes through agricultural lands, requiring caution to avoid contamination from spills. Respect for local wildlife and livestock crossings is important.

Communication Coverage

Mobile network coverage should be generally consistent along the National Highway, though rural segments may pose potential dead zones. Drivers should have communication backup plans, such as

satellite phones or offline maps.

Emergency Response Times

Given the proximity to small towns, emergency response times could be within 30-60 minutes. However, response could be delayed in more rural or less accessible segments due to narrow roads and traffic.

Overall Risk Assessment

The primary risks involve weather-related incidents, road quality variations, and traffic congestion during peak hours. Adequate preparation, cautious driving, and regular vehicle checks are crucial. Keeping informed about local regulations and remaining aware of alternative routes will help mitigate these risks.

Overall, the route is manageable for heavy vehicles, but careful attention to weather forecasts, road conditions, and alternative plans is essential for safe passage.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453